Imprisonment Trends and Prison Capacity in Europe:

A Comparative Regional Analysis (2014-2023)

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Image source: Business Standard – “NHRC notice to Kerala DG Prisons over lack of help for inmates’ education”.

Research Questions

Primary Research Question:

Do European imprisonment systems converge or diverge over time, and how do **structural**, **regional**, and **COVID-19-related factors** shape these dynamics (**2014-2023**)?

Secondary Research Questions:

1. **Trend Analysis:** How has **official prison capacity** evolved across **European countries** between **2014 and 2023,** and which countries exhibit the **largest relative increases** or **decreases**?
2. **Regional Patterns in Incarceration:** What **regional patterns** emerge in **imprisonment rates** across **Europe** and how do **Northern**, **Southern**, **Eastern**, and **Western Europe** differ in their imprisonment trajectories over time?
3. **Impact of COVID-19 on Prison Populations:** How did **prison populations** change across **Europe** during the **COVID-19 period (2020-2022)** and how did **incarceration trends** differ across **European regions** during **COVID-19**?

About Data

Dataset: *Prison capacity and number of persons held*

- ***Source:*** *Eurostat (official EU statistics) (view data)*
- **Structure used:** Country × Year (annual time series)
- **Indicators:** Official prison capacity (number), Official prison capacity (per 100,000 inhabitants), Actual prison population (number), Actual prison population (per 100,000 inhabitants)
- **License (data):** EU Open Data License – reuse and modification allowed with attribution
- **Data issues:** Some missing country-year entries; no prison-level detail

Methodology

- **Data Collection:** Official European prison statistics (2014–2023) were manually downloaded from public sources to ensure transparency and data reliability.
- **Data Processing & Cleaning:** Data were processed in Python using **pandas and numpy**, with checks for missing values, reporting gaps, and inconsistent country coverage across years.
- **Data Transformation:** Yearly datasets were merged into a panel structure, prison population **rates as reported in the original sources** were used, and regional and temporal aggregates were computed.
- **Analytical Techniques:** Descriptive statistics, trend analysis, regional comparison, COVID-period comparison, and outlier identification were applied.
- **Visualization Tools: Python (Matplotlib, Seaborn, Plotly)** and **Microsoft Excel** were used for data visualization.
- **Use of AI Tools:** ChatGPT was used for code refinement and visualization support, while all data handling and calculations were researcher-controlled.
- **Reproducibility:** All analysis code is publicly available on **GitHub (view code)**, enabling full replication using the same data sources.

Insights from the Data

Europe does not follow a single prison trajectory.

- Incarceration ranges from **34.9 per 100,000 in Iceland** to **403.0 per 100,000 in Türkiye**, with an **average of 124**, revealing **strong inequality across countries**.

- Over time, **national paths clearly diverge**: some countries **expand capacity** (**Türkiye +81.45%**), while others **reduce it** (**Lithuania -28.6%**).

- These differences **cluster regionally**: **Northern Europe** remains **low and stable**, while **Eastern Europe** concentrates the **highest levels and largest changes**.

- **COVID-19 exposed these structures rather than equalizing them**: prison populations **fell sharply** in some countries (**Kosovo -22.12%**) but **rose** in others (**Liechtenstein +27.27%**).

Countries with Rapid Prison Capacity Expansion

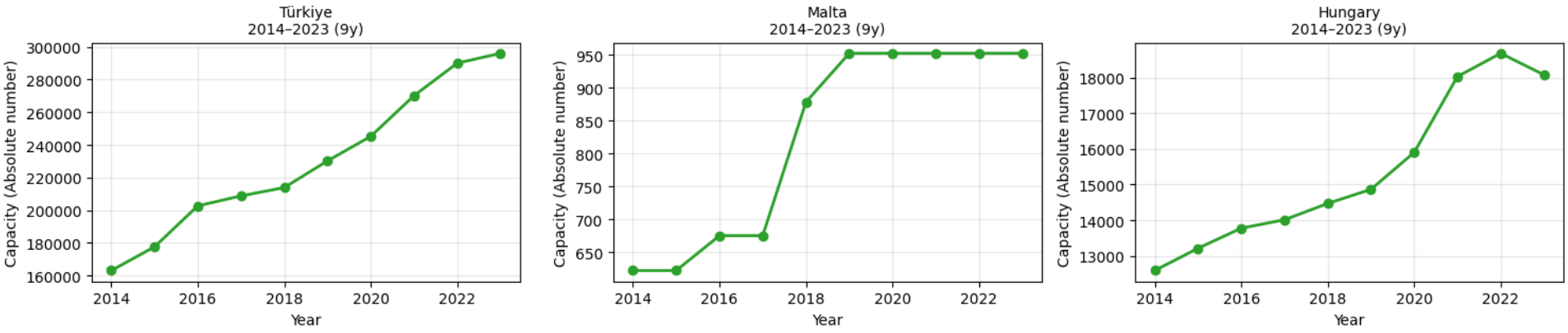

Figure 1: Countries with the Largest Increases in Official Prison Capacity (2014–2023)

Prison capacity trends across Europe are not uniform but follow clearly differentiated paths.

A group of countries shows **sustained expansion**, indicating **long-term growth in prison capacity** *(Top 3: Türkiye, Malta, Hungary)*.

Countries with Sustained Prison Capacity Decline

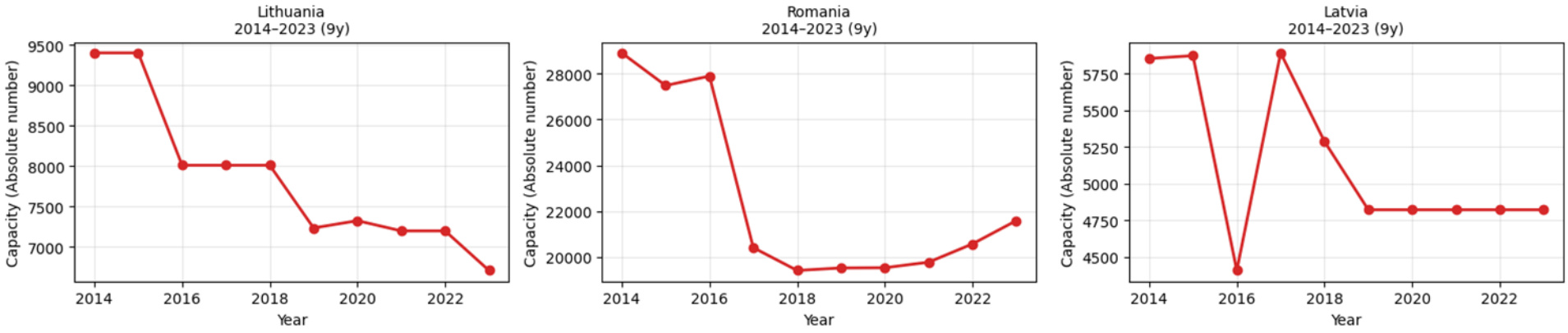

Figure 2: Countries with the Largest Decreases in Official Prison Capacity (2014–2023)

Another group exhibits **persistent contraction**, reflecting a **steady reduction in prison infrastructure** (*Top 3: Lithuania, Romania, Latvia*).

Countries with Stable Prison Capacity across Europe

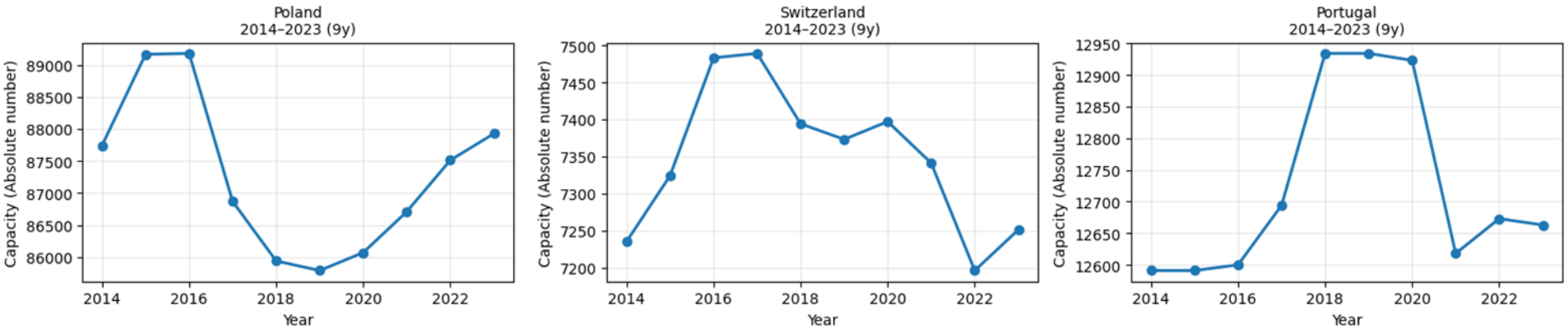

Figure 3: Countries with Stable Official Prison Capacity Levels (2014–2023)

Some countries show **no sustained long-term increase or decrease**, remaining broadly stable despite short-term fluctuations..

Together, these patterns highlight **divergent national trajectories** in prison capacity evolution.

Unequal Changes in Prison Capacity across Europe

Relative changes (measured as the percentage difference between 2014 and 2023) **in prison capacity differ substantially across European countries.**

Some countries exhibit **large proportional increases**, while others experience **significant reductions**.

Several countries show **minimal relative change**, indicating overall stability.

The ranked comparison in **Figure 4** highlights **heterogeneous national trajectories** over the study period.

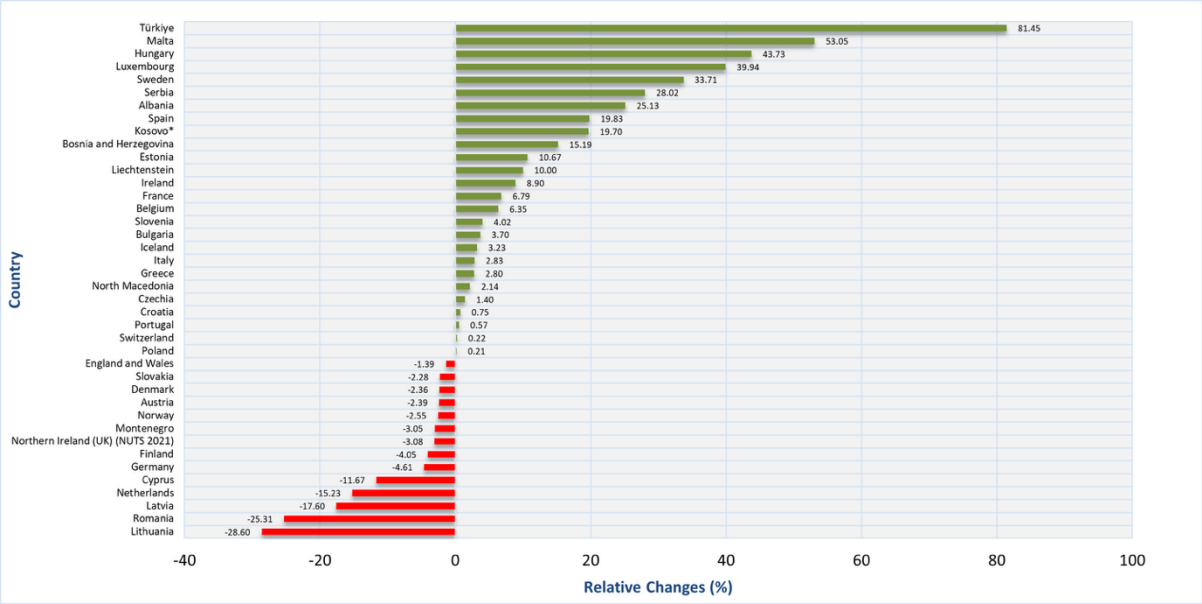

Figure 4: Relative Change in Official Prison Capacity Across European Countries (2014–2023)

Grouping Europe into Four Macro-Regions

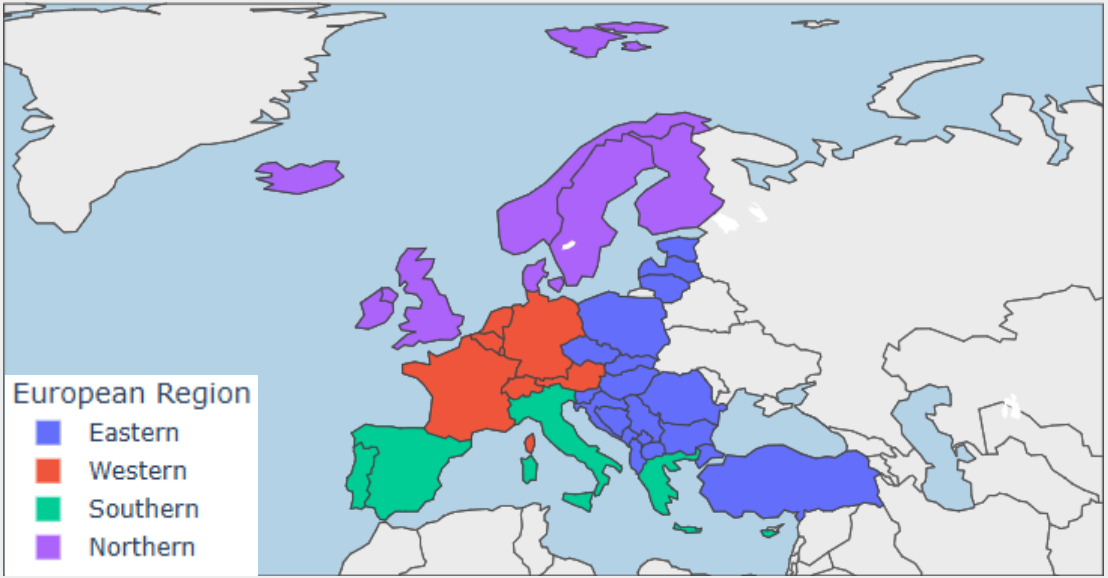

Figure 5: Geographical Classification of European Countries into Northern, Western, Southern, and Eastern Regions

Europe is often discussed as a single legal and political space.

But beneath this surface, **incarceration practices may follow regional logics rather than national exceptions.**

To move beyond country-by-country comparisons, Europe is grouped into **four macro-regions: Northern, Western, Southern, and Eastern.**

Distinct Regional Incarceration Trajectories

European incarceration patterns are clearly regional in nature.

Northern Europe is low and stable, Eastern Europe is high and uneven, while **Western and Southern Europe occupy intermediate but distinct positions.**

Incarceration in Europe is therefore **structured by region, not randomness.**

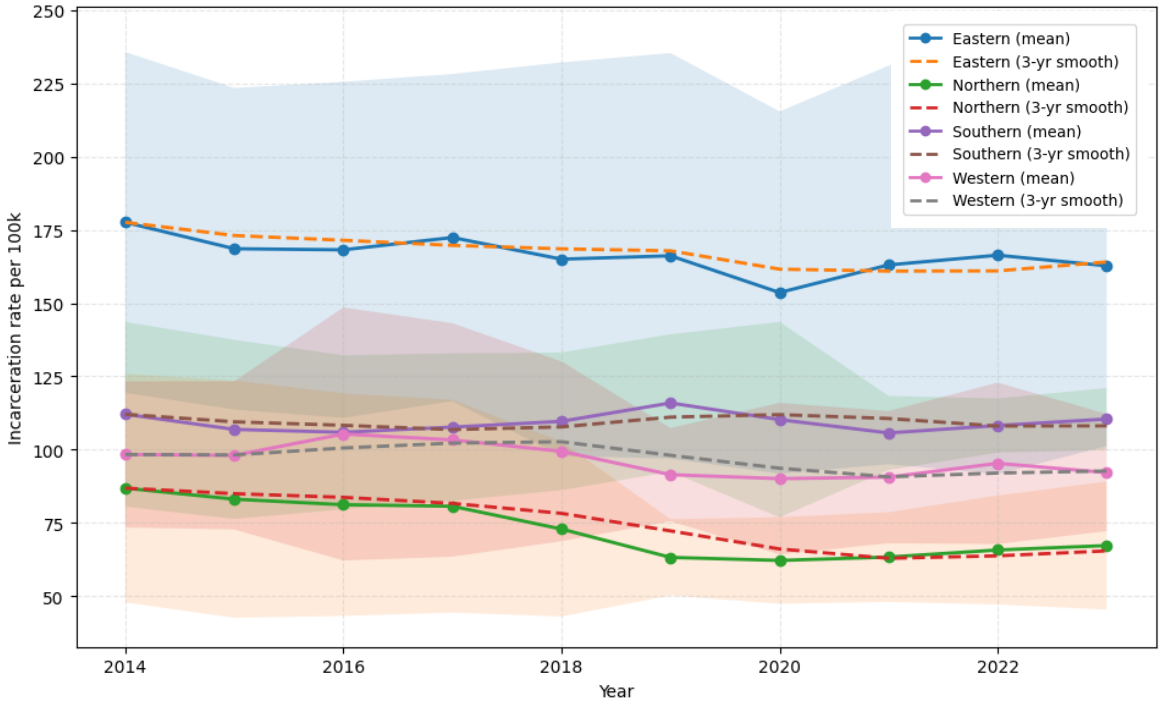

Figure 6: Regional Trends in Incarceration Rates Across Europe (average value) (Per 100,000 Population)

Hidden Inequality within European Regions

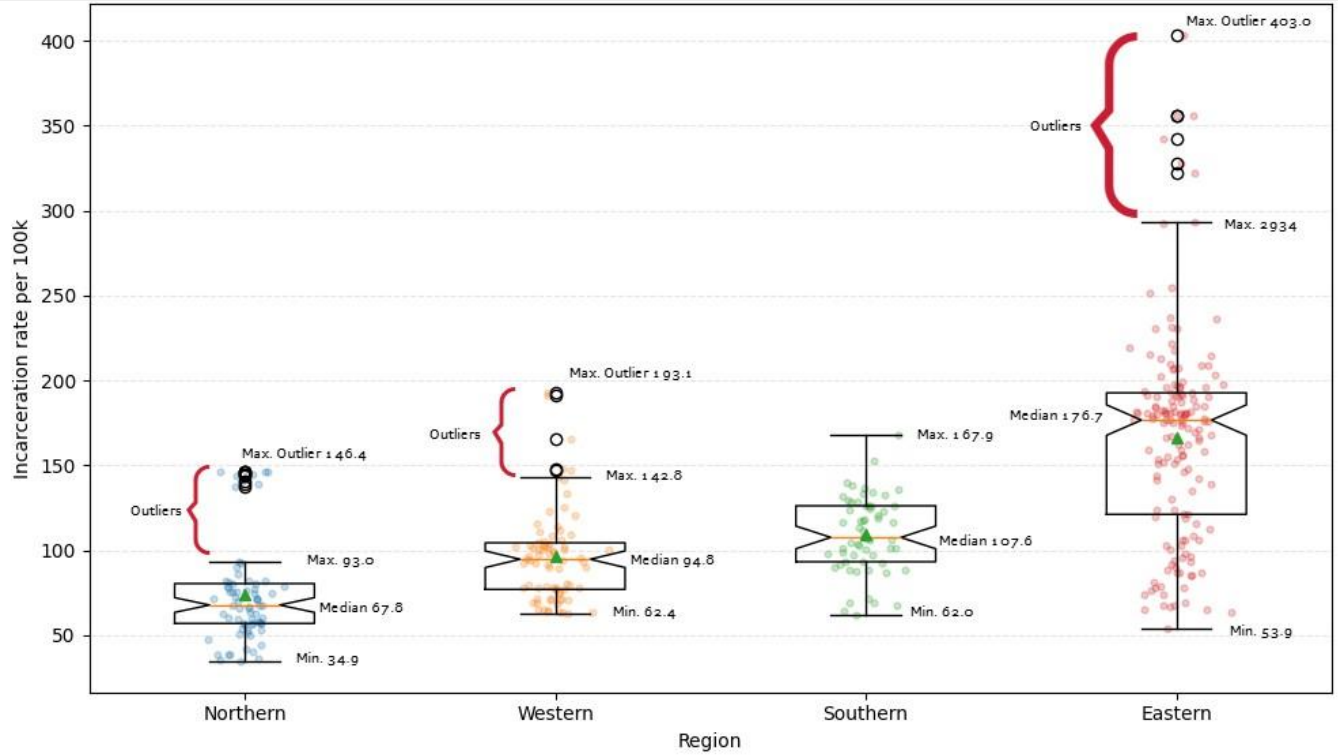

Figure 7: Cross-Regional Distribution of Incarceration Rates in Europe (Per 100,000 Population)

Trend lines show overall direction but hide differences within regions.

Northern Europe exhibits uniformly low incarceration rates, reflecting strong internal consistency.

Western and Southern Europe follow comparable trajectories, but with **noticeable internal variation.**

Eastern Europe departs sharply from this pattern, combining the **highest incarceration levels** with **substantial inequality across countries.**

Divergent National Responses during COVID-19

Prison populations across Europe did not respond **uniformly** during the **COVID-19 period**.

While **several countries** reduced **incarceration levels**, others showed **limited change** or **increases**, highlighting substantial **divergence in national policy responses** to the same **external shock**.

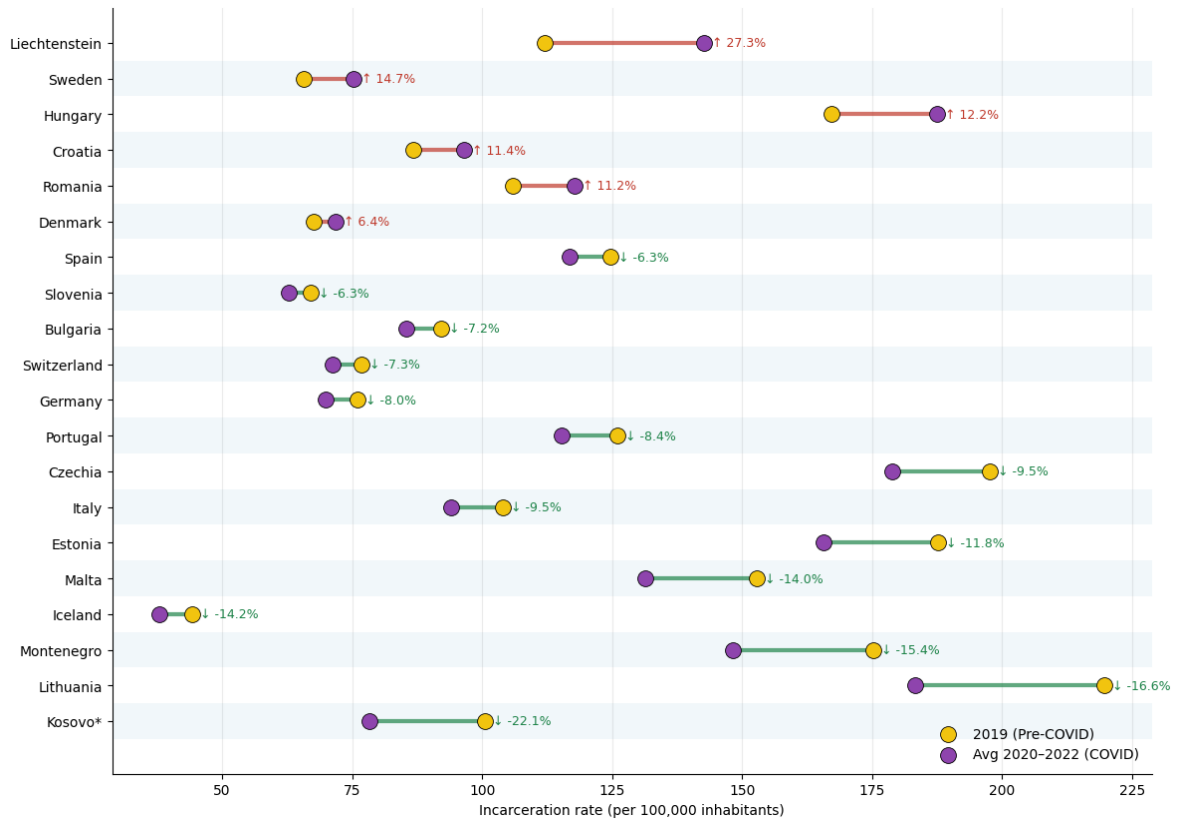

Figure 8: Country-Level Changes in Incarceration Rates Before and During the COVID-19 Period (2019 vs. 2020–2022)

Uneven Regional Effects of the COVID-19 Period

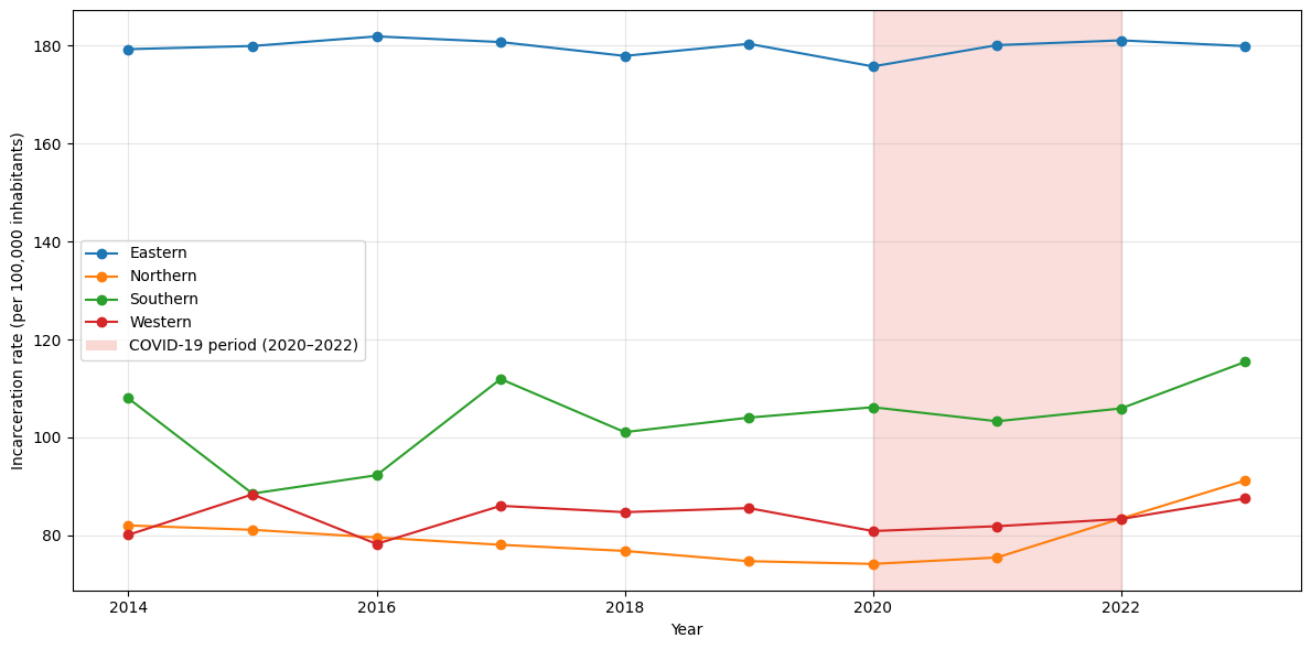

Figure 9: Regional Trends in Incarceration Rates Across Europe, 2014–2023

Regional incarceration trends indicate that the **COVID-19 period** affected **European regions unevenly.**

While **some regions** experienced **clear downward shifts**, others remained **stable** or **rebounded quickly**, reflecting **divergent regional trajectories.**

LICENCE

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ABOUT THE AUTHOR

MD. SHAKHAWAT HOSSEN

I am an aspiring Data Scientist currently pursuing an MSc in Data Science at the University of Milano-Bicocca. I am passionate about extracting insights, identifying trends, and implementing data-driven strategies to address practical challenges. My goal is to strengthen my analytical and technical expertise to deliver effective data solutions.

Technical Skills

Python | SQL | Excel | Data Analysis | Data Visualization | Statistical Analysis | Machine Learning